IoT & AI for Assisted Living

UNIVERSITY OF LIVERPOOL

Mia Hornett (201551655) Daniel Mitchell (201540943) Alvaro Mesa Giner (201656665) Luis Lozano Guillot (201514204)

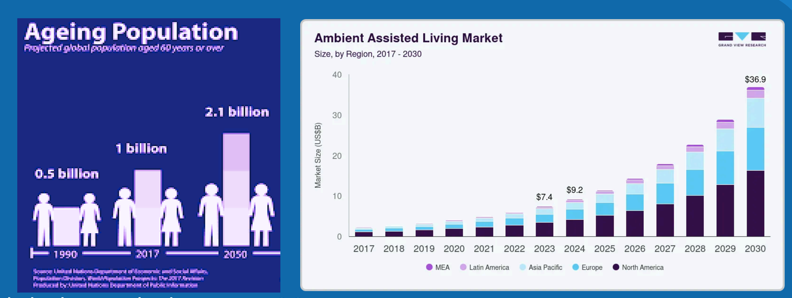
Supervisor - Dr Valerio Selis Assessor - Dr Junqing Zhang

Aims & Objectives

1. Create an IoT and AI system to assisted living
2. 8 devices - 5 sensors 3 actuators
3. AI to learn routines and flag anomalies
4. Privacy first design
5. App for caregivers and family members
6. Continuous monitoring

Current Market

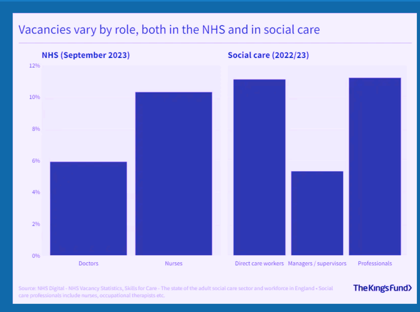
Market Demand



Global population over 60 reaching 2.1billion by 2050 [1].

Ambient Assisted Living Market $9.2 Billion USD (2024) [2]

Work Force Crsis



Staff shortages
131,000 vacancies in the UK social care sector [3]

Existing Products

Smart Home Healthcare devices

Healtcare Specific — Adapted Generic Devices

SECOM [4] StackCare [5] Apple Home[6] Alexa Together[7]

Commerical Edge

Highly scalable
One off costing model
Individual homes or large housing associations
App for caregivers
AI Routine Learning

Background Theory

Passive sensing

Sensors collect data without requiring interaction.

PIR, Pressure, mmWave, Water, Light

Edge computing

Processing occurs locally on the Raspberry Pi

Reduce latency, Privacy protection, Lower cloud dependency

Behavioural Pattern Learning

AI learns daily routines from sensor activity patterns to detect

Deviations, Anomalies, Unusual inactivity

Human in the Loop Learning

Caregivers can manually input routines through the app to improve AI predictions & manage alerts & anaomolies

Project Plan

Deliverables

1. Scalable Hardware Prototype
2. Edge AI Pipelines
3. Cloud Infrastrucure
4. Caregiver Mobile App
5. System Documentation

Milestones

Semester 1

1. Hardware Archiecture & Design
2. Local Network Setup
3. Initial System Build

Semester 2

1. Complete System Building
2. Cloud System Deployment
3. App Integration

MIA: Hardware Development; Feature Extraction

DAN: Backend & Architecture Development; Backend & Frontend Integration

ALVARO: AI & Routine Learning

LUIS: App design and implementation

MQTT Network; Smart Home; Anomaly Alert Mapping

Method

Mia

Hardware Selection & Integration

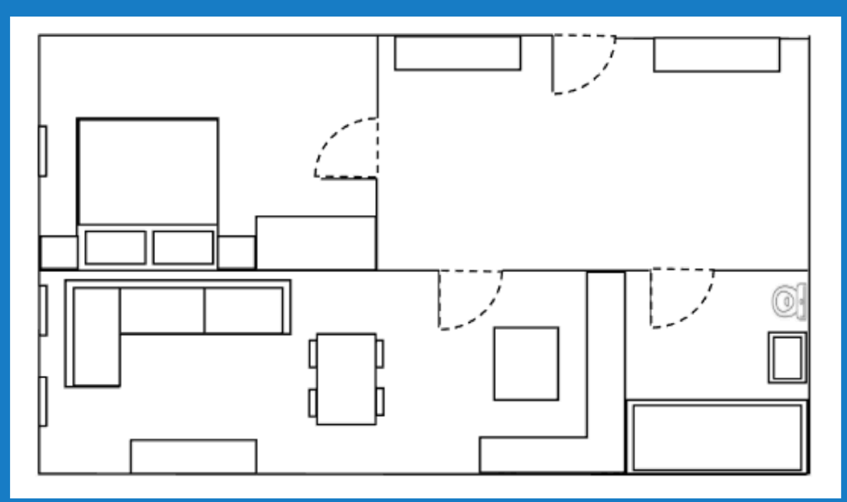
- Sensor & Actuator Selection
- Circuit Design & Prototyping
- Circuit Testing

Firmware Developement

- Scripts for ESP32 & Sensor/Actuator Developed using Arduino IDE
- MQTT & Secure Port 8883 Transmission
- Certificates & Keys Generated

Physical Protyping

- PCB Design
- CAD
- Final Product Assmebly



Dan

System Architecture Design

Local Edge Network — Cloud Network

Edge System Development and Implementation

- Local Mosquitto Broker Installation
- Python Edge Service
- Local Communication Testing

Backend Development and Implementation

- Cloud Backend Deployment
- Database and Notifications
- Cloud System Testing

Alvaro

Data Generation & Preparation

- Synthetic IoT sensor dataset generation
- Dataset structured for machine learning analysis
- Data preprocessing and feature preparation

Data Generation & Preparation

- Python development in Visual Studio Code
- Isolation Forest algorithm implementation
- Model training for anomaly detection

Anomaly Detection Testing

- Detection of abnormal sensor patterns
- Integration with smart home monitoring system
- Evaluation of model prediction results

Luis

UI Development

- Flutter cross-platform app
- Sensors / Routines / Settings interface

Secure communication

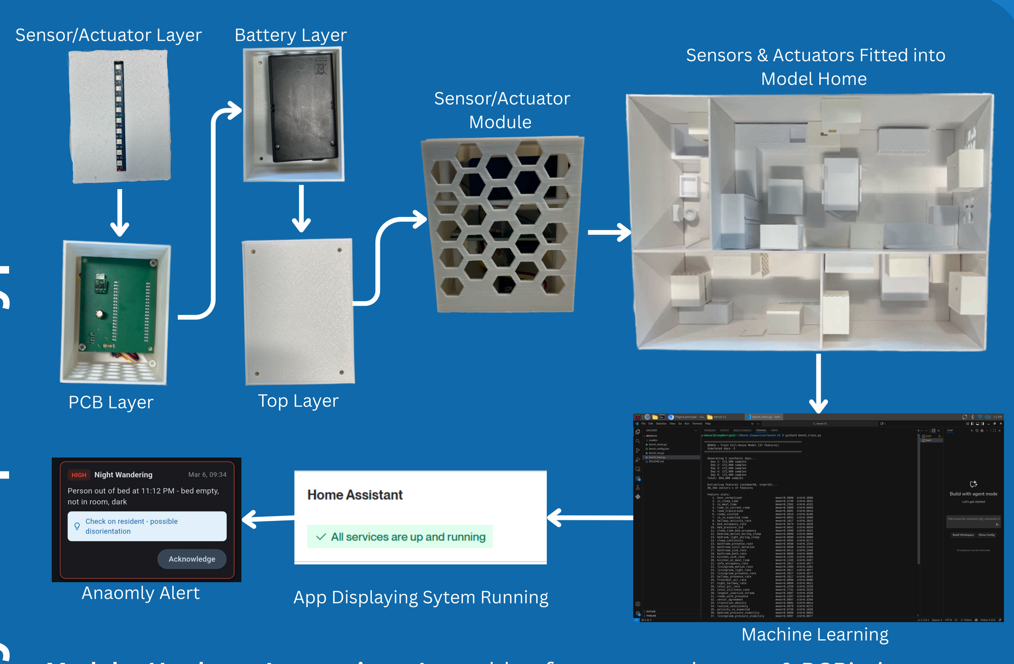
- HTTPS connection to FastAPI backend
- Data retrieved from cloud database

Routine input System

- Supports AI learning pipeline
- Caregivers can manually define routines

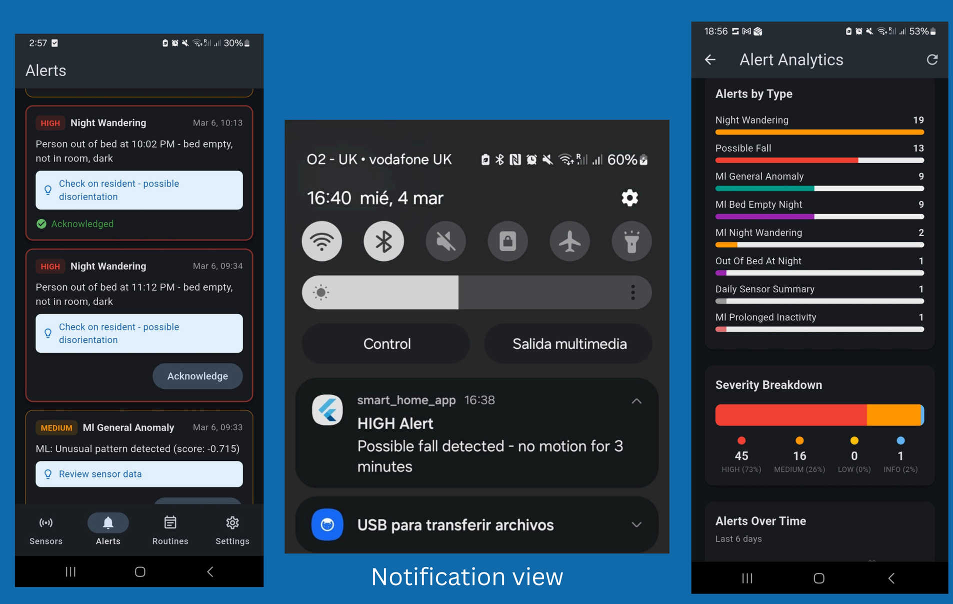
Results & Discussion

System prototype



Modular Hardware Integration - Assembly of custom enclosures & PCB's into compact deployable modules proving physical system viability
Scaled Environement Viadilation - 1:75 scaled model allowed for controlled sptial testing, local MQTT communiction and data injestion prior to full scale trails
End to End System Functionality - Validation of entire archicture, physical environemtal triggers process by ML model to generate real time anomaly alerts in the app

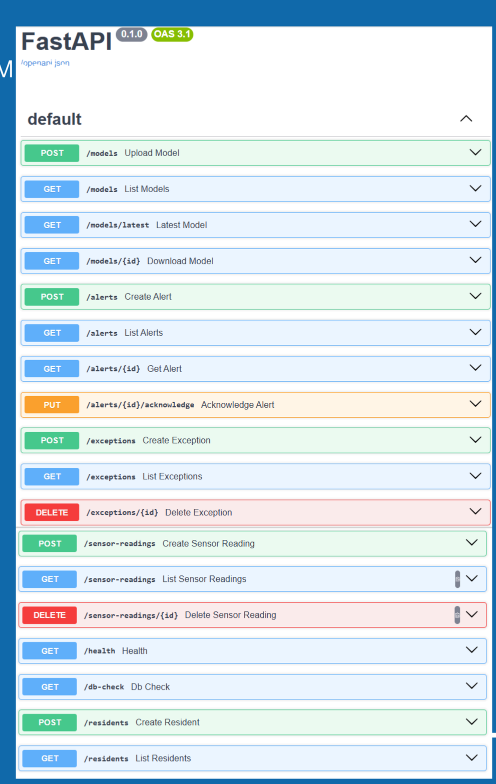
Mobile Application Prototype



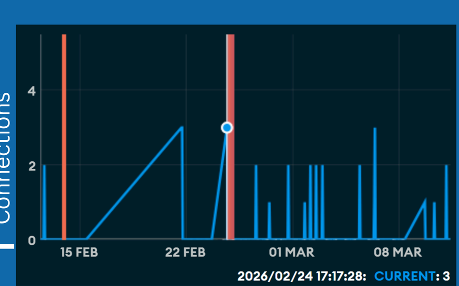
Alerts recieved — Histogram

Actionable Edge AI Insights - Translaton of complex models into clear alerts for care givers
Real Time Push Notifications - Instantly alerts carers to anomalies
Secure Behavioural Analytics - Display historical alerts to give a view of long term bhevaioural trends

Backend implementation



Swagger UI - Endpoints utilisations



Database usage

MongoDB stores resident credentials, routine backups, anomalies
Uses **FCM** (Firebase Cloud Messaging) for alerts
Secure connection requires **authentication** via app keys

FastAPI endpoints accessed via the frontend and machine learning models continuously
Uses secure **HTTPs** address
Accesses and edits data saved in **Cloud Database**

Conclusion

Outcomes

Architectural Success - Deliverable a highly scalable, privacy preserving hybrid IoT & AI platform
Hardware & Edge Reliability - Passive sensing & Actuator array achieve secure low latency data transmission.
IIntelligent Processing – Sensor streams are analysed using an Isolation Forest anomaly detection model to identify abnormal behavioural patterns.
Remote accessibility - Deployment of FastAPI cloud infrastructure & cross-platform Flutter application, gives reliable offline access to criitcal data.

Future Work

Expand deployment - support multiple homes and care facilities.
Integrate additional IoT sensors and personalised ML models.
Scale the cloud architecture to support thousands of devices.
Real-Time analytics - Allows 24hr access for caregiver/ family
Advance system to allow for 'plug & play' external IOT devices

References

[1]United Nations, "World Population Ageing," United Nations, 2017. Available: https://www.un.org/en/development/desa/population/publications/pdf/ageing/WPA2017_Highlights.pdf [2]Grand View Research, "Ambient Assisted Living Market Size & Share Report, 2030," www.grandviewresearch.com, 2023. https://www.grandviewresearch.com/industry-analysis/ambient-assisted-living-market-report [3]S. Mallorie, "The Adult Social Care Workforce In A Nutshell | The King's Fund," The King's Fund, May 02, 2024. https://www.kingsfund.org.uk/insight-and-analysis/data-and-charts/social-care-workforce-nutshell [4]SECOM, "Wellness | Assisted Home Care | SECOM Plc," SECOM Security Systems, Mar. 31, 2025. https://secom.plc.uk/smart-wellness/ (accessed Nov. 12, 2025). [5]StackCare, "StackCare," StackCare, 2025. https://www.stackcare.co.uk/ (accessed Nov. 12, 2025).[6]Amazon, "Alexa Together launches to help customers remotely care for loved ones," US About Amazon, Dec. 07, 2021. https://www.aboutamazon.com/news/devices/alexa-together-launches-to-help-customers-remotely-care-for-loved-ones[7]Apple, "Home App," Apple (United Kingdom), Sep. 29, 2025. https://www.apple.com/uk/home-app/